

Project Initialization and Planning Phase

Date	7 th July 2026
Team ID	
Project Title	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To automate credit card approval decisions using machine learning techniques and improve decision speed and accuracy.
Scope	The project predicts applicant creditworthiness using financial and demographic features and provides instant approval recommendations.
Problem Statement	
Description	Traditional approval systems are time-consuming and may lead to inconsistent decisions and financial losses due to risky applicants.
Impact	The proposed solution reduces processing time, improves consistency, minimizes financial risks, and enhances customer experience.

Proposed Solution	
Approach	The system preprocesses applicant information, extracts useful features, and uses machine learning algorithms to predict approval status.
Key Features	• Automated approval prediction • Real-time decision making • Risk analysis • User-friendly web interface • High prediction accuracy



	- Real-time decision-making for quicker card approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Model training	Google Colab CPU/GPU
Memory	RAM specifications	12 GB
Storage	Dataset and models	20 GB
Software		
Frameworks	Python frameworks	Scikit-learn

Libraries	Additional libraries	scikit-learn, pandas, numpy , matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle Credit Card Approval Dataset, csv, 438,557 application records and 1,048,575 credit records